

Digital Converged Triple Play Service

hanarotelecom

June 3, 2005

I . What is TPS?

II . TPS Cases

III. Beyond TPS

IV. hanarotelecom's TPS Strategy



I . What is TPS?

II . TPS Cases

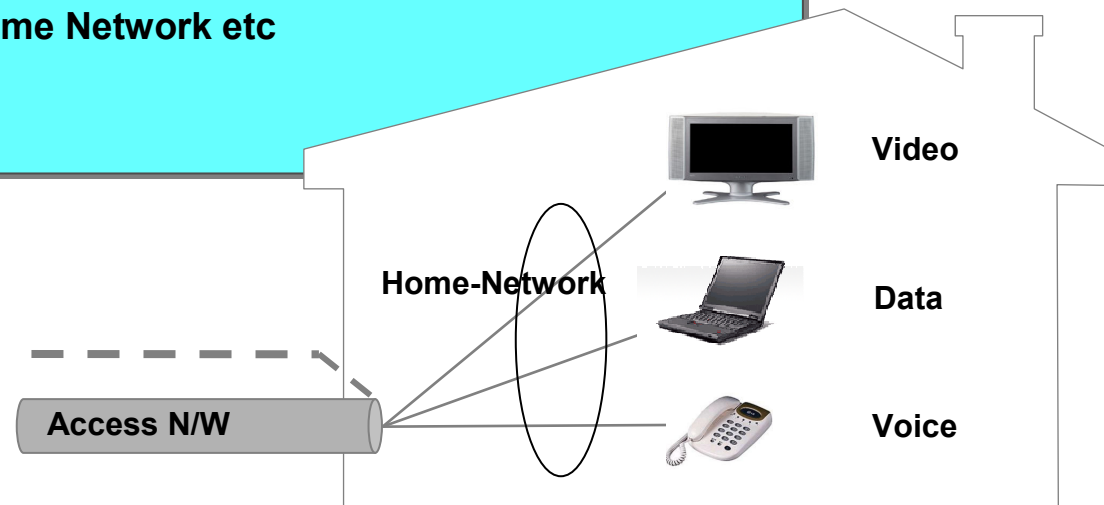
III. Beyond TPS

IV. hanarotelecom's TPS Strategy

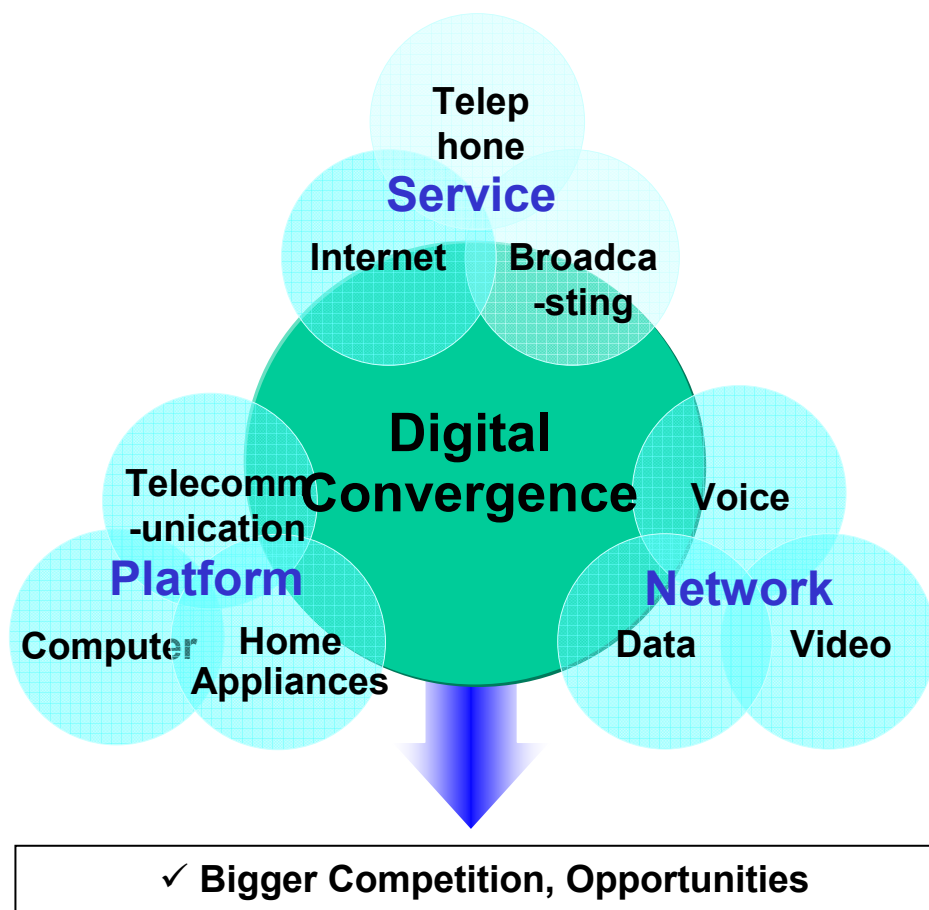
I . What is TPS?

Data, Voice and Video service through Single or Heterogeneous Network.

- High speed internet : HFC, xDSL, Ethernet, FTTH
- Voice : PSTN, VoIP
- Video : CATV, DBS, Broadband TV
- Converged Service : VOD, DP, Home Network etc



I . What is TPS?



Drivers	
Technology	• Video, Data Processing Storage • Intelligent Network • Converged platform/Terminal
Regulation	• Demand for regulation change • Convergence in business : telco, CATV, Broadcast -> open competition
Market	• One Stop Service • Needs, customer group

👉 Accelerate the changes : M&A, Strategic Alliances

Network Integration (BcN)

Regulatory issues

I . What is TPS?

Various TPS

Network	Single Access N/W	Heterogeneous Access N/W
	• Wire Access N/W - xDSL+ PSTN/VoIP + BB-TV - HFC + VoIP + CATV • Wireless Access N/W - Mobile (Voice) +Data + Video - PDA Phone (Voice) +Data + Video	• Hetro Wire Access N/W - Telco Wire (Data+Voice) + CATV/DBS(Video) • Mobile Access - Wibro + Mobile - DMB + Mobile
Service Provider	Single Service Vendor	Multi Vendors
	• Vertical Integration of service and platform - Common Service Framework	• Products Bundle : Sales Channel • Strategic Alliance : complimentary services



I . What is TPS?

II. **TPS Cases**

III. Beyond TPS

IV. hanarotelecom's TPS Strategy

II. TPS cases

Domestic Market

Service Provider	TPS Service
hanarotelecom	• Joint service package with SO : Data + VoIP + CATV • Hanafos + Skylife • IPTV in lab trial
KT	• KT + SkyLife ; Bundle products • Home-N • Videotelephony
CATV	• MSO : Digital Media Center - Digital Cable + VoIP(ASP) + Data



Your Friend in the Digital Age.



[Login to view/pay your bill](#)

[LOGIN](#)

[Need password?](#)

[Forgot password?](#)

[For Home](#)

[For Business](#)

[Order Services](#)

[View / Pay Bill](#)

[Customer Support](#)

[About Cox](#)

[Find service in your area](#)

[Zip code](#)

For Home

▶ Bundled services

Check out our
Special Offers
on Cox Services

EasyPay
An easier way to
pay your Cox Bill.

Bundled Services



- ▶ Exclusive savings
- ▶ One simple package

▶ Cox can simplify everything and deliver all your services on one simple bill. You can't get that with satellite.



Learn more about the Cox Digital Suite
by entering your zip code below.

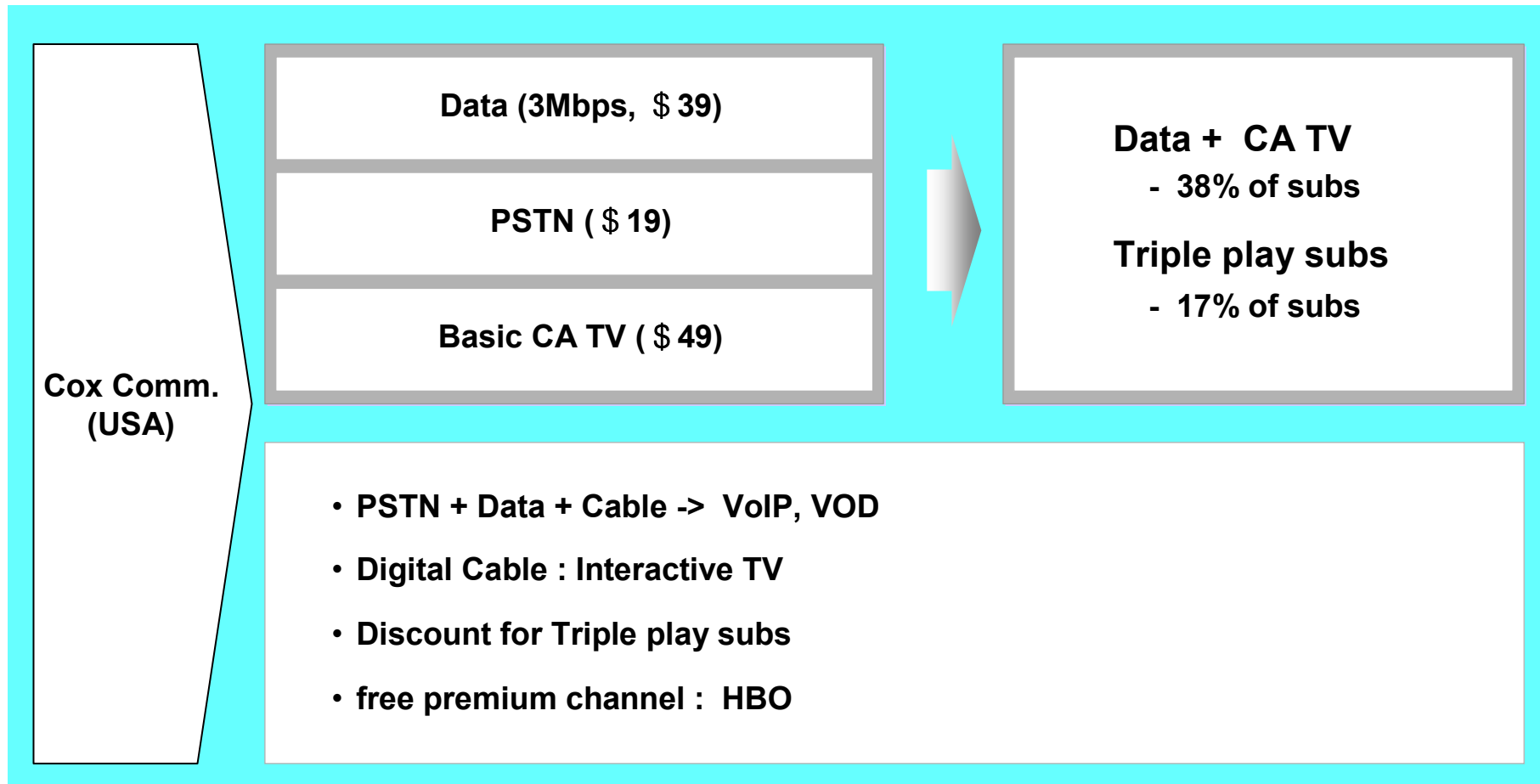
Check for special offers in your area!

Enter ZIP code: click go

[GO](#)

[SPECIAL OFFERS](#)

[ORDER ONLINE](#)



FASTWEB☐ **OFFERTA FAMIGLIA**

RAI CLICK

ONTV

FASTWEB BASE

VOCE

MEGA INTERNET

TV DI FASTWEB

PAY TV

CANALI TEMATICI

INTERNET ED E-MAIL

VIDEOREC

**VERIFICA DISPONIBILITA'**

Sei già raggiunto
dalla rete FASTWEB?
Clicca QUI per scoprirlo!

Mappa dei servizi
per la Famiglia

**IL TUO PREVENTIVO****FASTWEB Base**

FASTWEB Base 25,00€

↳ 19 euro per un anno -6,00€

**Voce**[Visita la sezione](#)**Mega Internet**[Visita la sezione](#)**TV di FASTWEB**[Visita la sezione](#)**Servizi Aggiuntivi**[Visita la sezione](#)**Importo Mensile Totale*:**

19,00€

ONTv

Entra nel mondo della TV on Demand: con **ONTv** scegli liberamente cosa guardare, quando lo desideri, tra oltre 1.400 titoli.

**ONTv****5,90 Euro/mese**

Aggiungi al preventivo



ONTv è libertà di visione. Sei tu che crei la TV per te e la tua famiglia scegliendo tra un'offerta di programmi continuamente aggiornata.

Con **ONTv** hai sempre a disposizione una videoteca ricca di titoli emozionanti provenienti dai più importanti nomi dell'intrattenimento: 20th Century Fox, Universal Studios, Buena Vista International, Dreamworks, Medusa, Turner, BBC, Discovery Channel, Bim, Mikado e tanti altri.

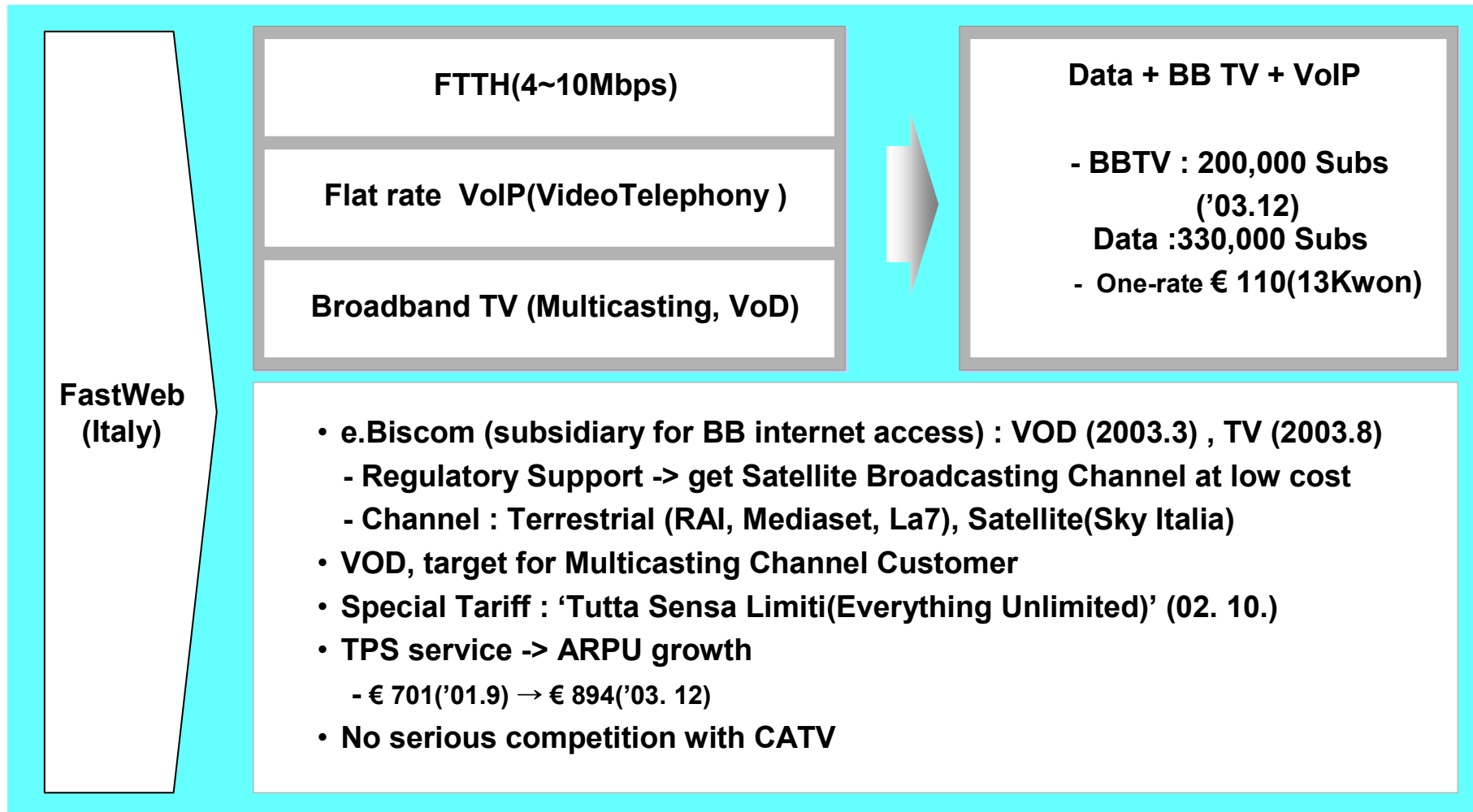
Entra nel mondo della TV on Demand: con **ONTv** scegli liberamente cosa guardare, quando lo desideri, tra oltre 1.400 titoli.

Sottoscrivendo **ONTv** puoi:

- accedere liberamente ad un'ampia proposta di **programmi per tutta la famiglia**: cartoni animati, sport, documentari, concerti, approfondimenti, magazine e tanto altro;
- acquistare i **film in Video on Demand** (prime visioni, grandi classici, disponibili anche con edizioni in Dolby Digital 5.1 e film on demand) oltre che

OFFERTA AZIENDA

COMPANY INFORMATION





BACK HOME

DATE: 24/05/2005

LOGIN 中文

nowChannels nowSchedule nowServices myAccount customerServices businessCustomer

now BROADBAND TV presents to you
the "Disney Channel Magicland"

Date : 28 - 29 May (Sat - Sun)
Time : 12:00 - 6:00 pm
Venue : **TOYS "R" US**,
G/F Ocean Terminal,
Tsim Sha Tsui



Exciting programs include :

- Children fun games with many prizes to give away
- Disney Channel's tri-lingual live broadcast inspires children in learning different languages
- Disney friends face-painting activity

Fabulous gifts* include :



Limited-edition Disney gifts and free Nestle cereals breakfast products* :



Available while stock lasts. © Disney

APPLY NOW !
Subscribe to **NETVIGATOR** BROADBAND Service and enjoy exclusive **now** Broadband TV service.
• \$0 monthly fee
• Free decoder
• Installation fee waiver



Win tickets to **Batman Begins** Advance Screening in HK

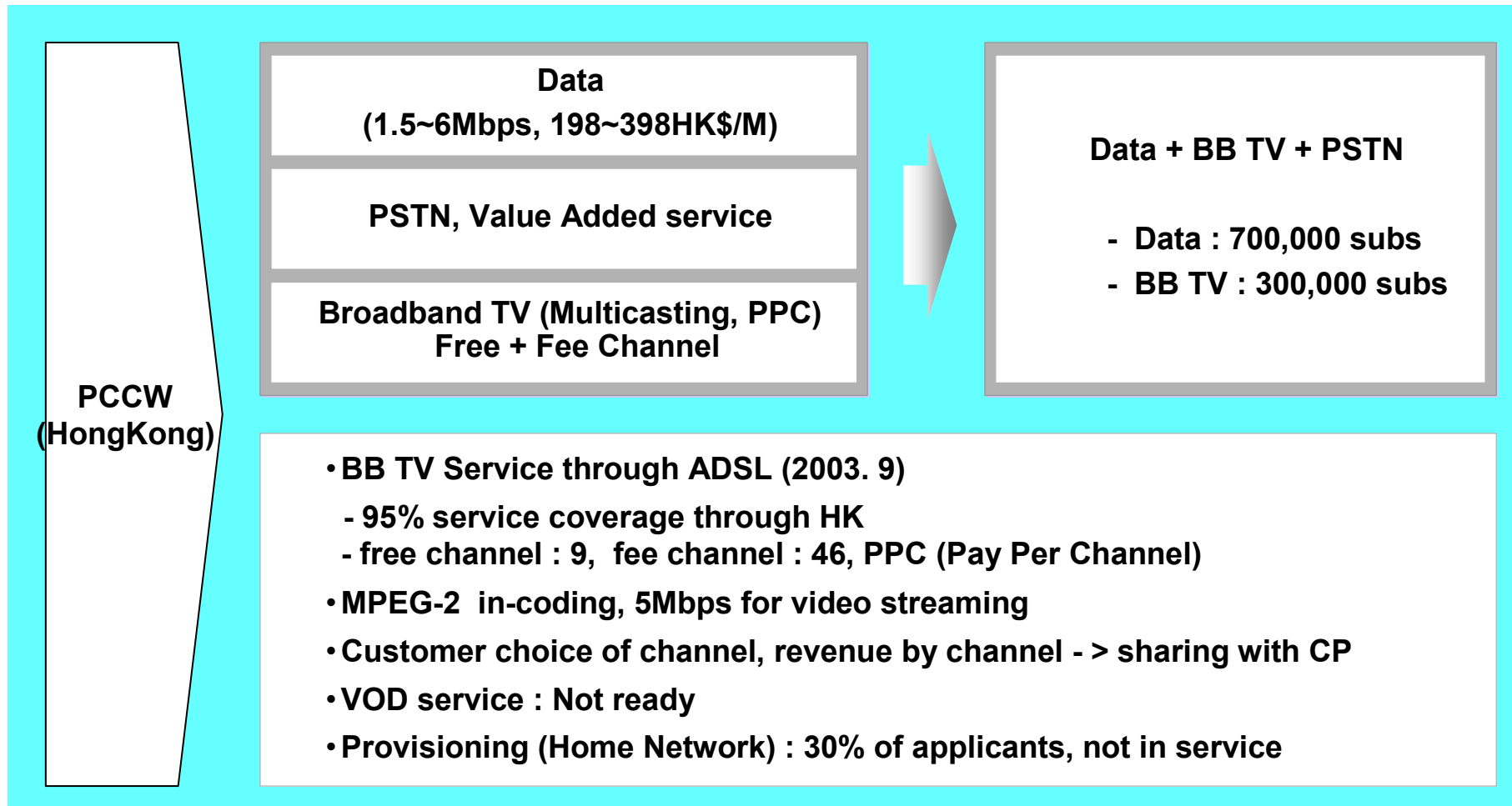
24 HOURS SALES HOTLINE
2888 0008
PRESS **1** TO APPLY

now Broadband TV
Amazing
Member Gets
Member Program



Commercial Users

Residential Users





[Yahoo! JAPAN](#) - [ヘルプ](#)

こんにちは、ゲストさん

[トップ](#) - [光サービス](#) - [ADSLサービス](#) - [サイトマップ](#) - [ログイン](#)

Yahoo! BB入会案内 - 光サービス

[光トップ](#)

[サービスタイプ](#)

[標準サービス](#)

[オプション](#)

[料金](#)

[お申し込み手順](#)

これからのYahoo! BB 光は、
インターネットとテレビが楽しめる。

Yahoo! BB 光 TV package 新登場



一戸建て住宅にお住まいの方

Yahoo! BB 光 新規お申し込み

パソコンからお申し込み

電話番号を入力して対応エリアを確認

市外局番 市内局番 下4ケタ

 - -

確認する

電話でお申し込み・ご相談

0120-33-4546

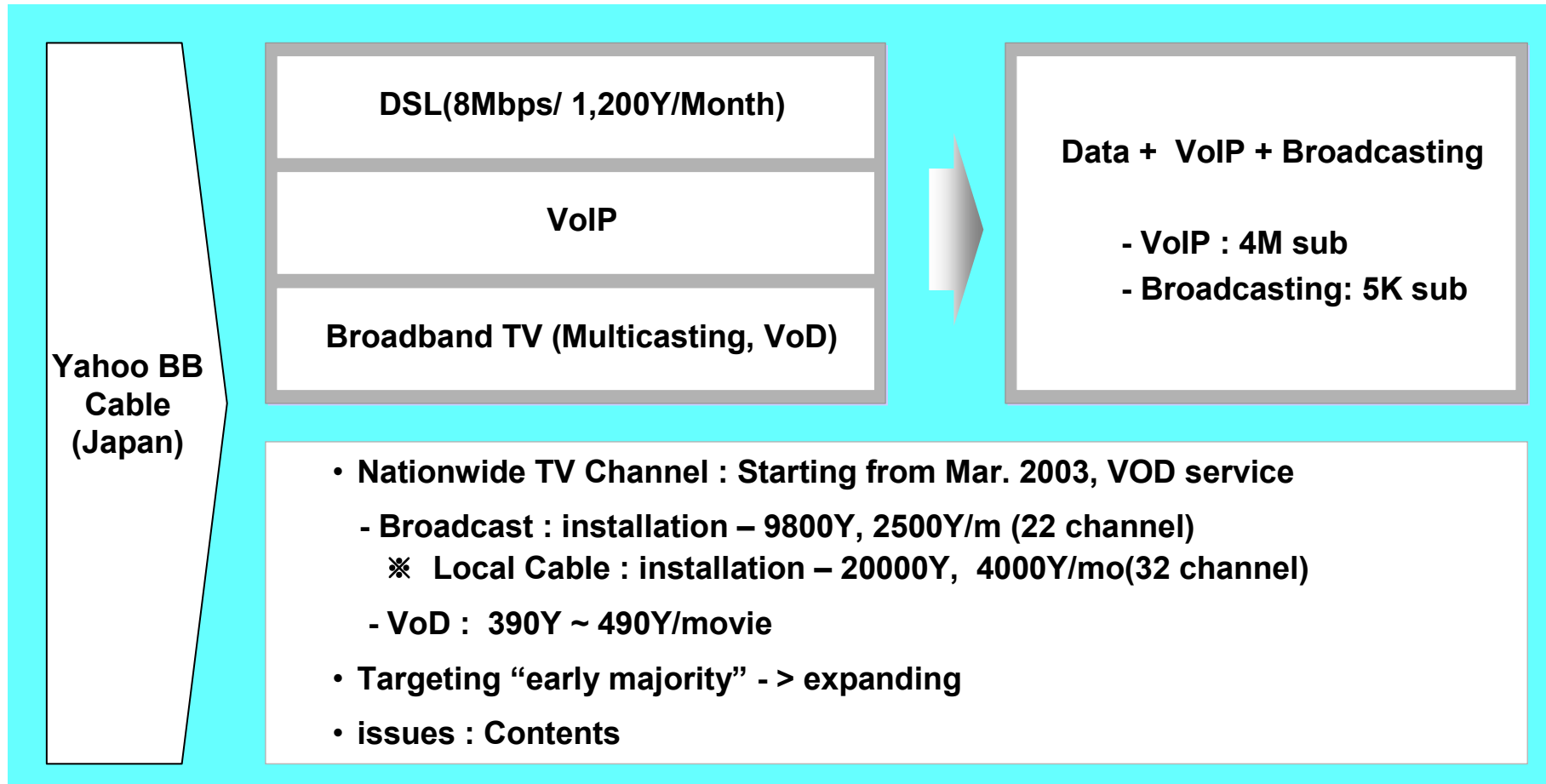
平日 9:00~22:00 土日祝 8:00~22:00

[Yahoo! BB 光へのサービス変更](#)

[Yahoo! BB会員限定
光にしよう! キャンペーン](#) **NEW!**

関連情報

- [Yahoo! BB 50M](#)
- 簡単にすばやくブロードバンド化
- [Yahoo!ニュース](#)
- 高速・常時接続
- [Yahoo!ショッピング](#)
- コンピュータ



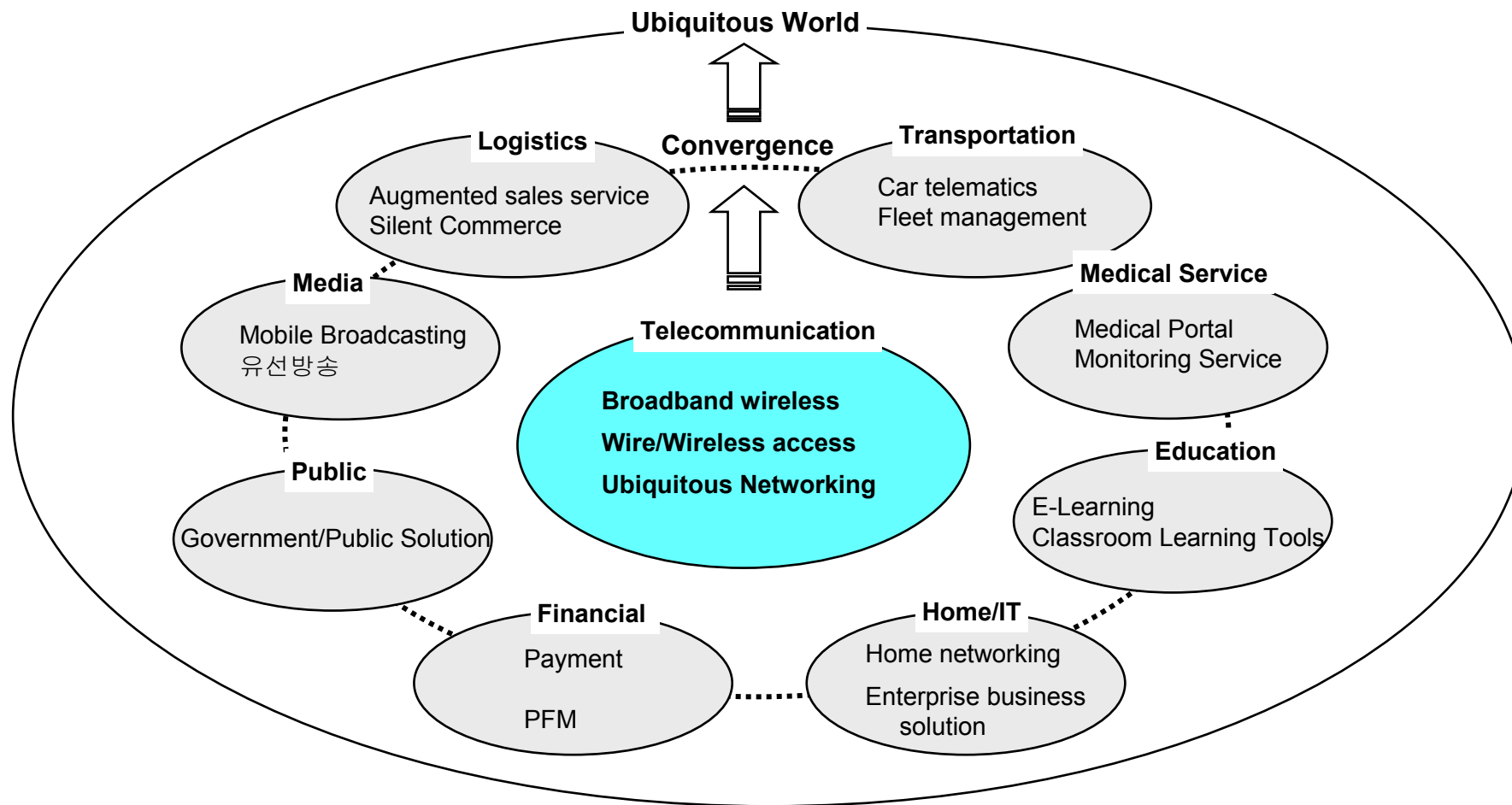


I . What is TPS?

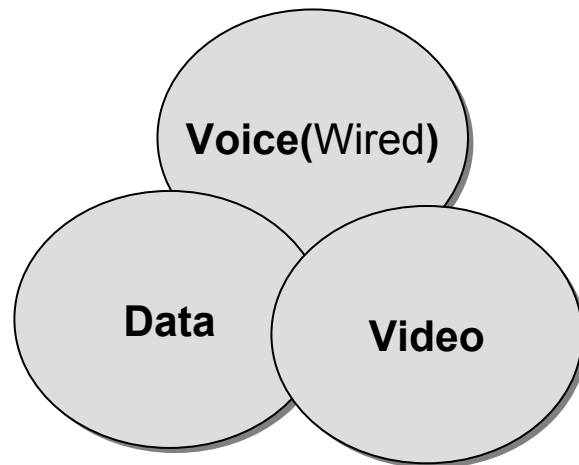
II . TPS Cases

III. Beyond TPS

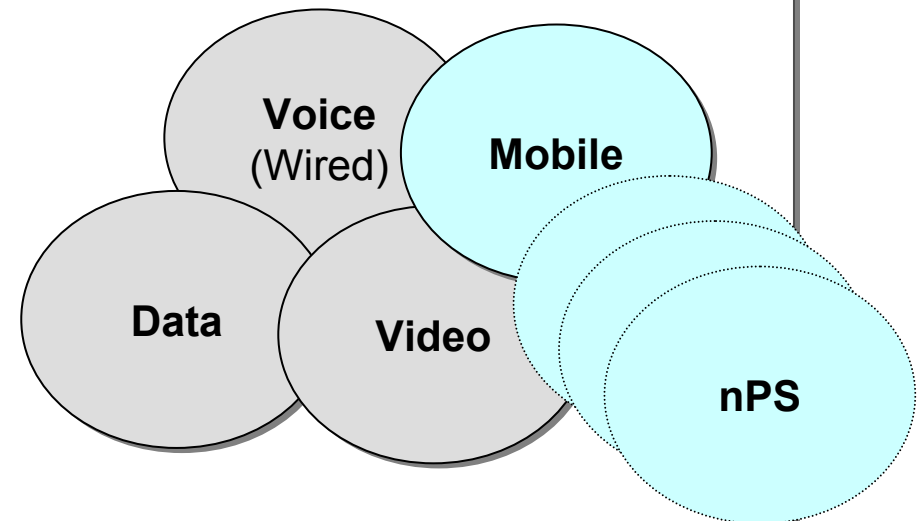
IV. Hanarotelecom's TPS Strategy



Triple Play Service



Quadruple Play Service



Drivers and Challenges	
Technology	• Multicasting capabilities in Network • Quality guaranteed QoS • Bandwidth • All-IP environment
Business	• Competitiveness in cost, quality • Killer services • Alliances between service providers
Regulation	• TV service on IP Network -> Regulation changed (Fair competition) • Legislative activities : Broadcasting Business Act, Service Act

I . What is TPS?

II . TPS Cases

III. Beyond TPS

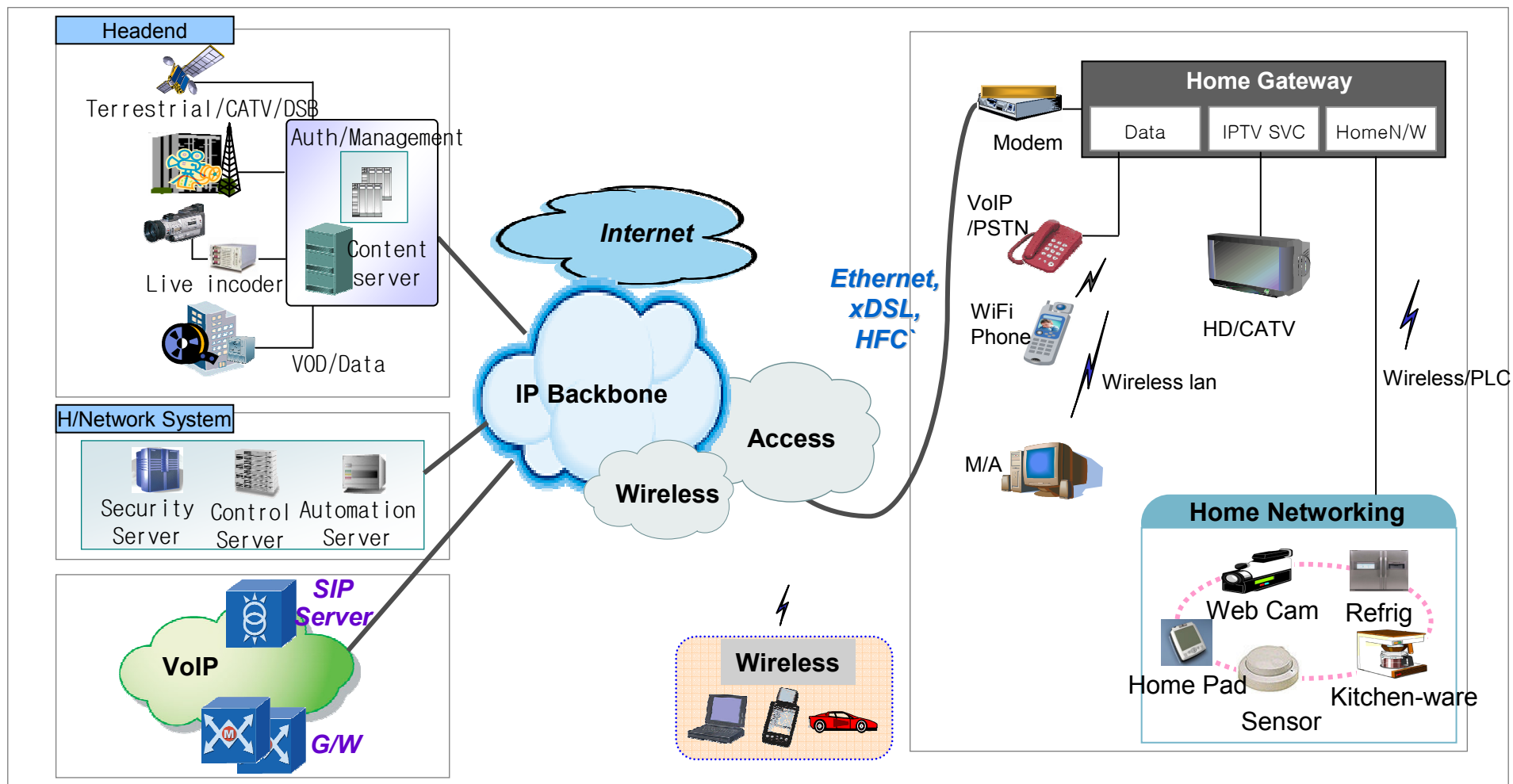
IV. **hanarotelecom's TPS Strategy**

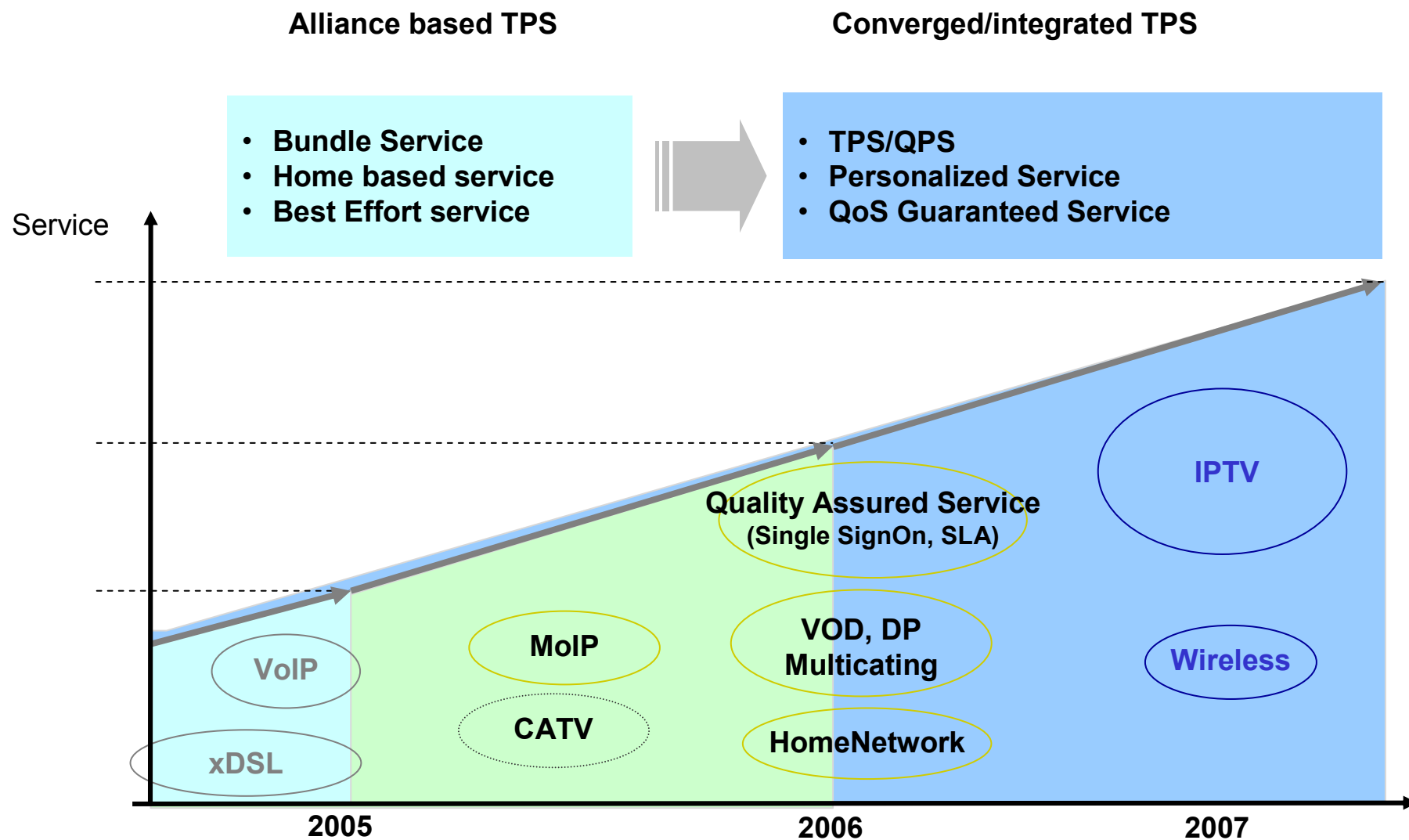
MoIP	• VoIP(Wi-Fi Phone), MoIP(QoS guaranteed videotelephony, MRBT, MCID, MMS, Video conference) • Content service for Video phone, Platform lease • Application service (distant learning, monitoring, commerce)
Data	• Multi-Line (Wireless Interface) • Traffic or Usage • Guaranteed Bandwidth • Mobility (Fixed mobile) • Various speed options
VAS	• Security (Vaccine, firewall) • Home-Viewer • Parental control • Access hour control • Realtime Video(Traffic, security)

VOD	• NVoD, RVoD on TV - high resolution(HD TV), Hi-fidelity voice(CD) -> Movie and eLearning content
Multicasting	• Video streaming through IP network. - Terrestrial TV, Pay Per View Channel - CP Multicasting • HD Multicasting
IPTV	• Weather, News, Traffic, Local community portal • TV portal : Internet portal on TV • T-Commerce : Home shopping • Entertainment - Game : download, online - Karaoke On Demand
Other	• PVR • TV-SMS • BB-TV lease

Home Automation	• Light, Curtain and Heater control • Home Appliances Control : mobile • Distant Meter Monitoring : Gas, Power, Water
Security	• Gas valve, fire alarm and surveillance • Door control by internet web cam , mobile • Video Monitor – Kids and pet care
Others	• Public service : local government services • Medical service

IV. hanarotelecom TPS Strategy TPS Picture





- Upgrade the service network (VDSL, Ethernet, FTTH)
- Existing customer focus : Retention rate
- Development of VAS, bundle package and loyalty program
- M&A , Strategic alliances
- Service/support quality focus

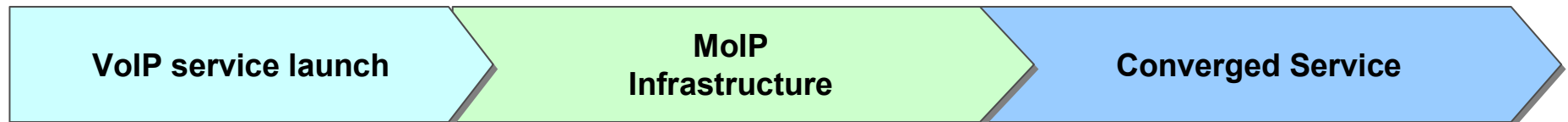
Network	Service Quality	Competition
• ADSL→ VDSL • Next generation CMTS • Replace the legacy network • Speed/quality management	• Leverage of existing resource • Customer Satisfaction • New service products	• Critical Mass : Strong 2nd player • Thrunet Acquisition • Alliance with SO

Key Factor	Strategy
• VoIP service - increase the coverage - regulatory environments • HFC(single house)→xDSL/LAN(APT)	• First, leading service player in VoIP services • IP based VAS : Video telephony

2004

2005

2006



- Digital voice
- Audio Ring back
- Text CID
- Videotelephony

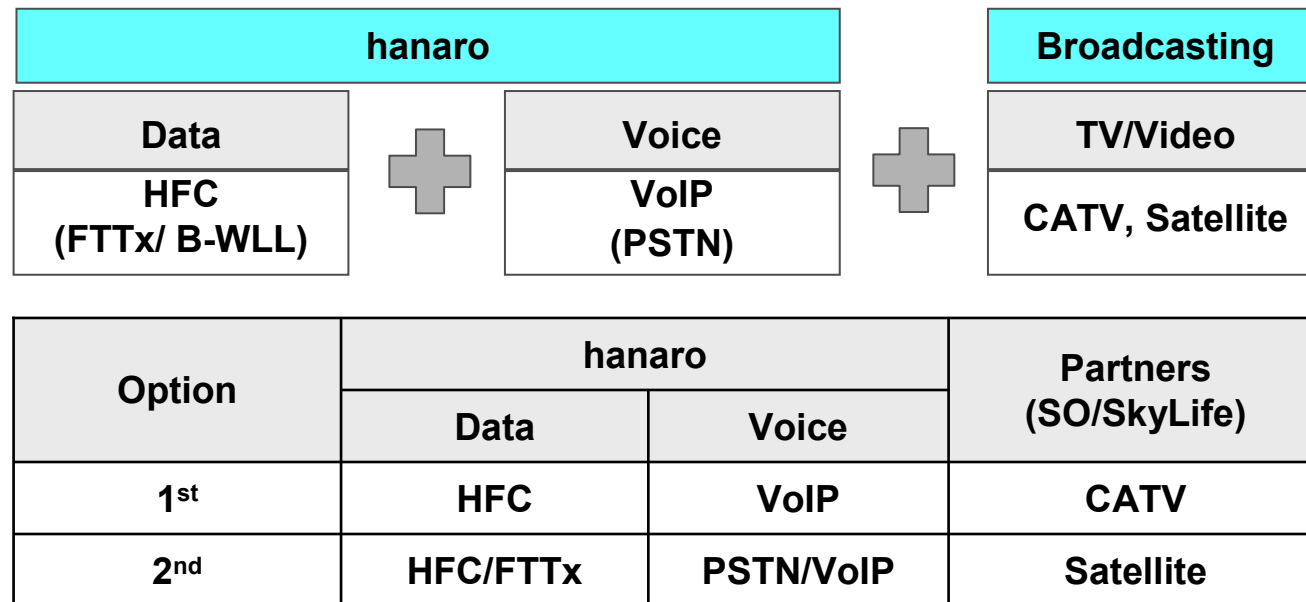
- Voice Conference
- MMS
- MRB/MCID
- Wi-Fi phone

- Platform
- Contents

Strategy

- Joint market development with SO
- Bundle Products -> provide customer with the value
- Area Marketing
- Hanaro's TPS (longterm) – IP TV

Service Products



IV. hanarotelecom TPS Strategy Home Network

Strategy

- Alliances with SKT, Electronics vendors, Constructions– risk sharing, service development
- Broadband TV -> Key services
- Target services development : focused on Apt.

2004

Launched joint trial services

- **Customer Needs**
 - Trial services : customer behaviors, market acceptance
- **Digital Home Biz Model**

2005

Trial services for commercial deployment

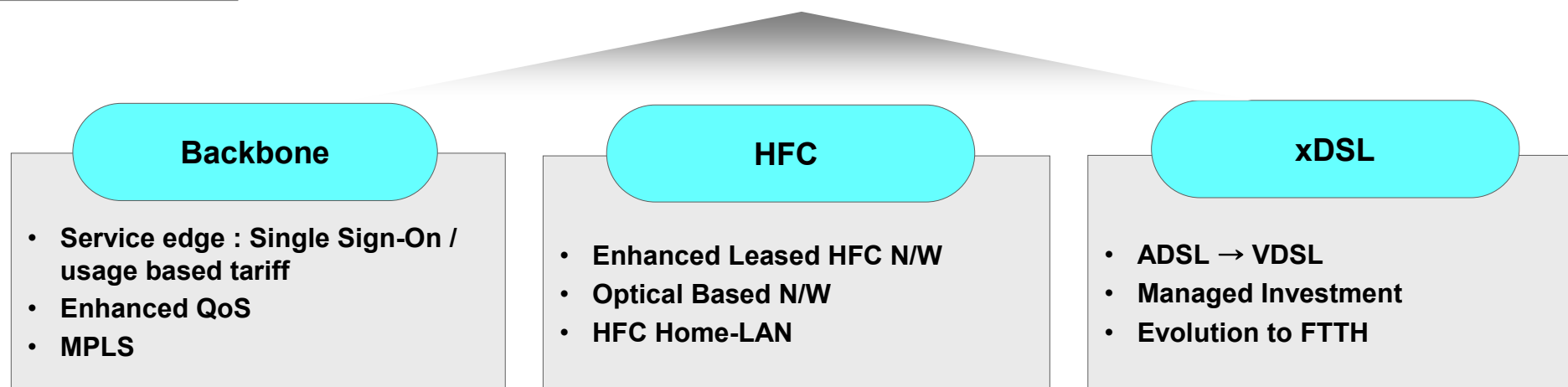
- **Wire/wireless + Home Entertainment**
- **Home gateway for Broadcasting + Data + Home network**

2006

Commercial Services

- **Converged services**
- **Intelligent Home gateway**

QoS	• Best Effort QoS → Guaranteed QoS. (separate standard : data, voice, video) • Traffic Control, Security & Mobility • Subs information management : Authentication
Multicasting	• Network Multicasting • Interactive Contents Consuming SVC
IPv6	• ALL-IP Network
other	• Asymmetric Network → symmetric Network • Mobility



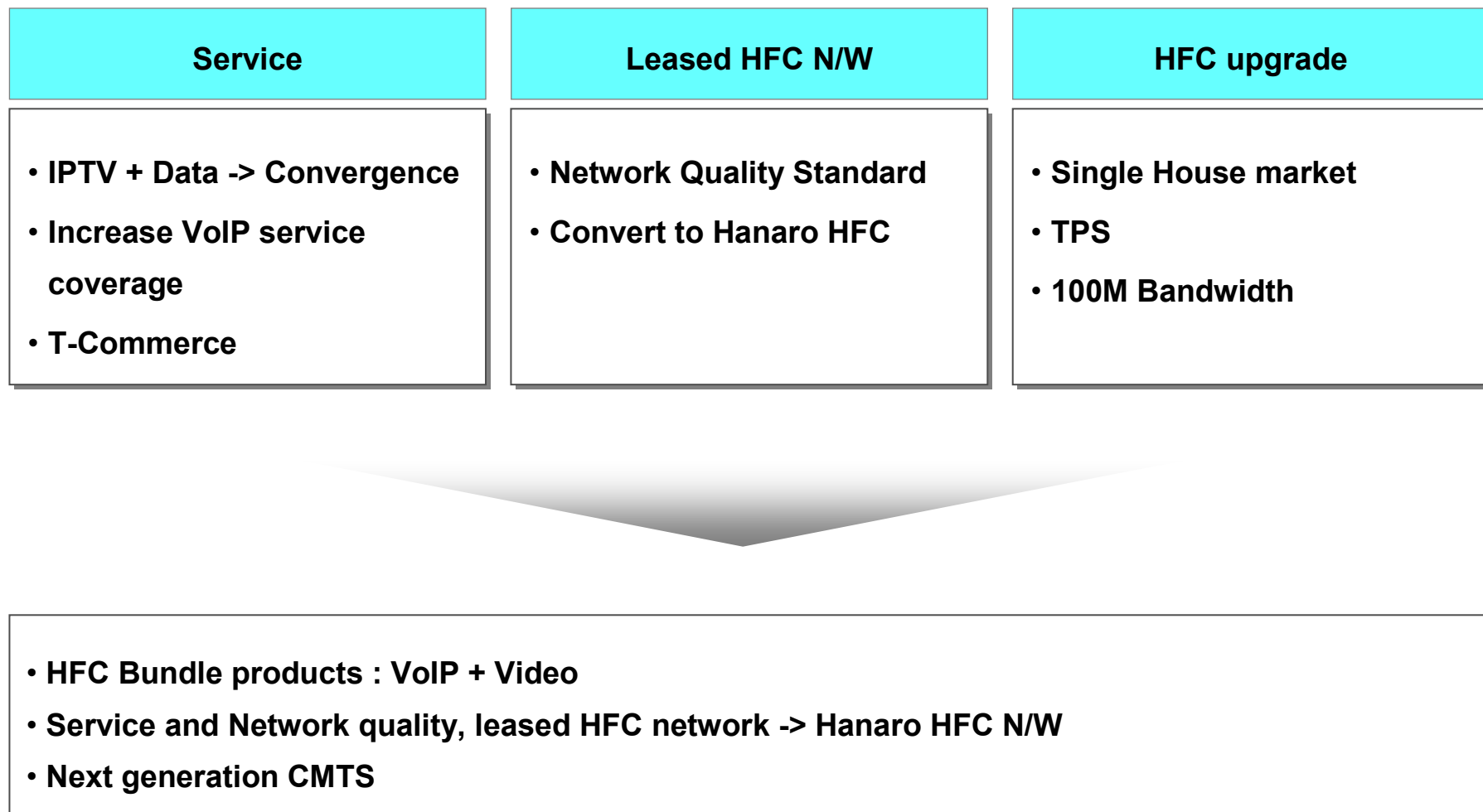
Increased Traffic in existing network

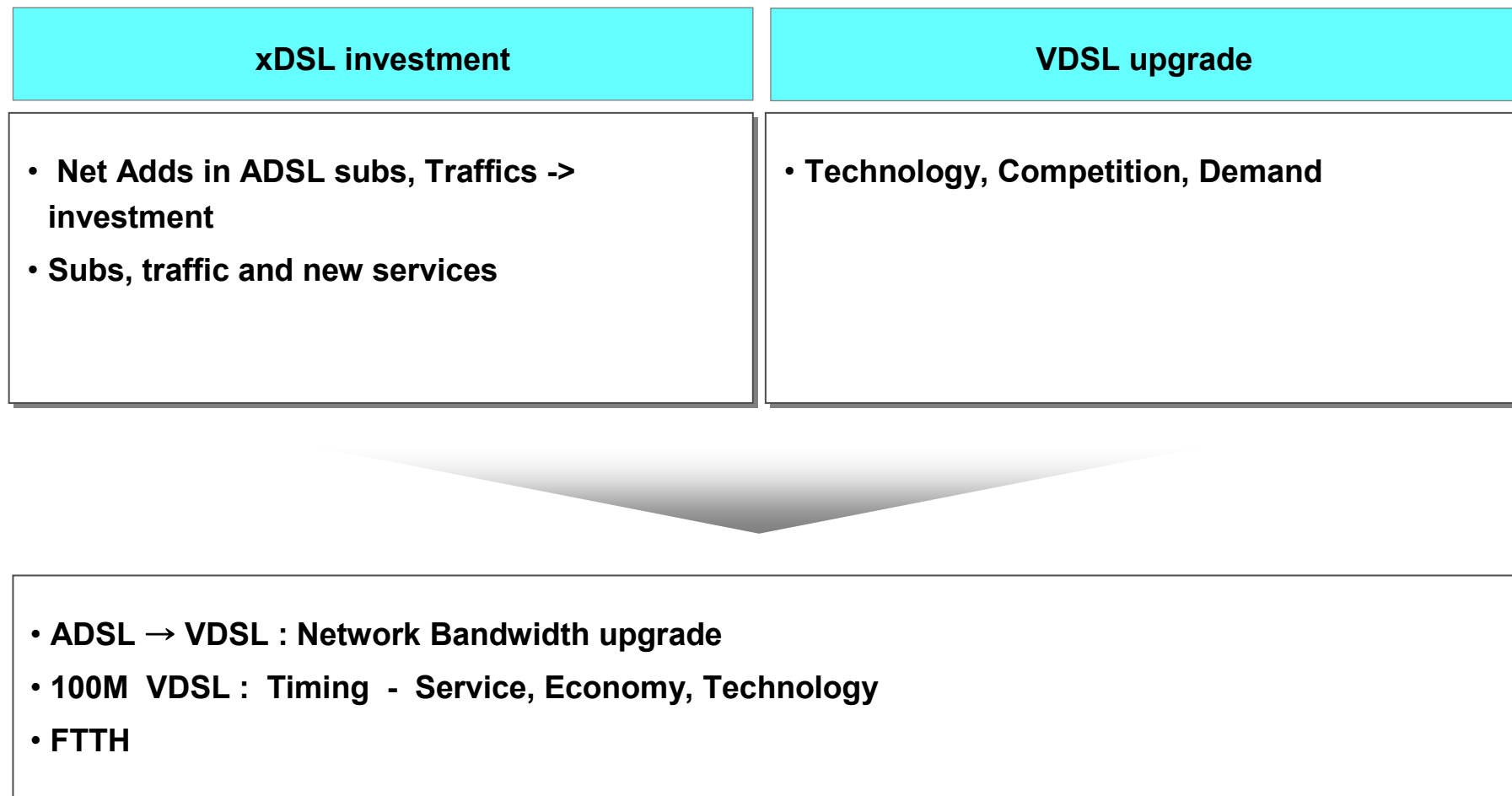
- Fast increase in Traffic
 - '03 – '08 (4.3 times, 40%/년)
- Increase traffic causes more investment than net adds in subs.

Quality assured service

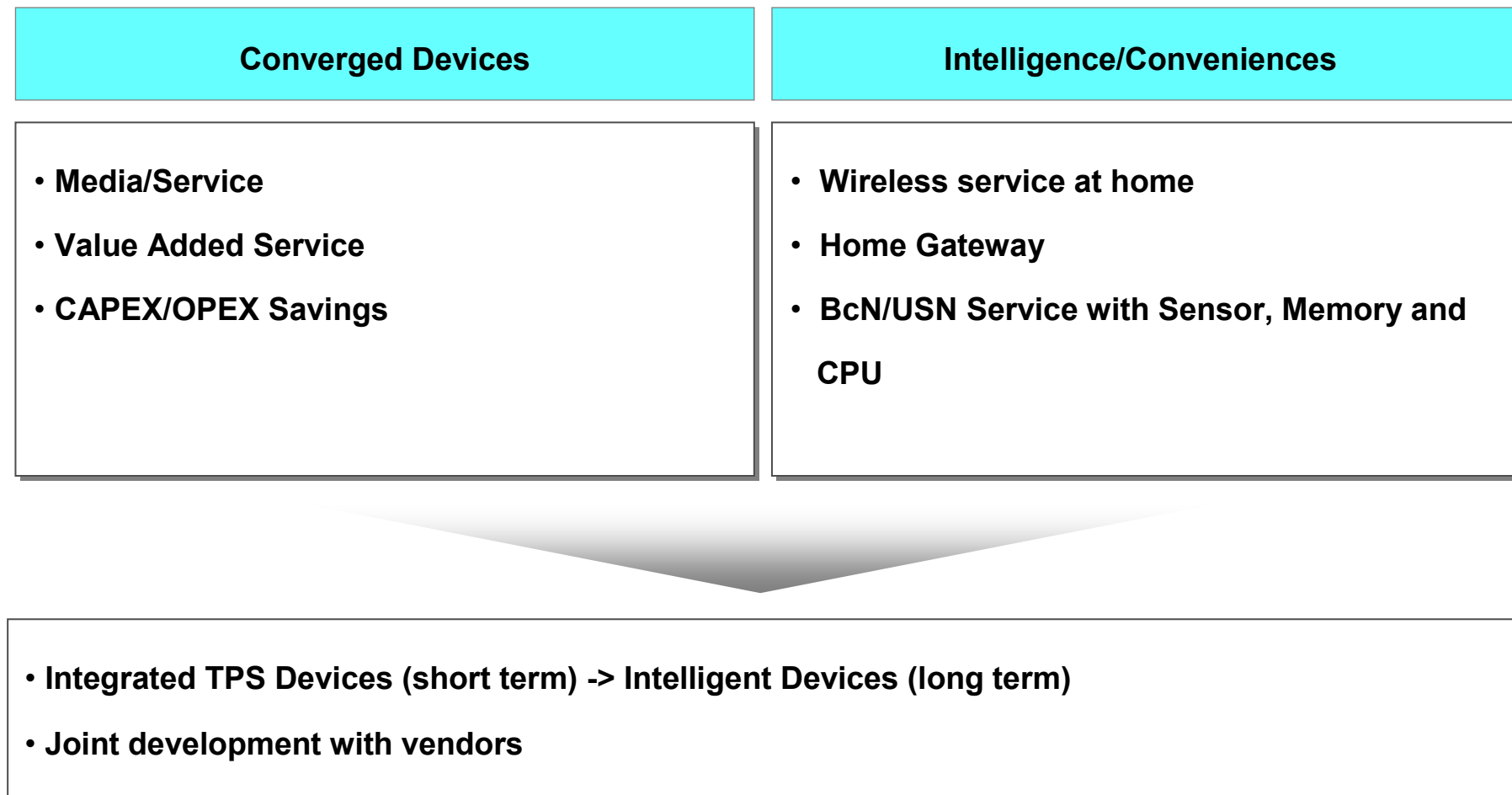
- VoIP/MoIP/VoD,BBTV
- Existing Backbone : Best effort network

- MPLS network for Quality Assured Services : VoIP/MoIP, VoD
- Service edge : traffic management





IV. Hanarotelecom's TPS Strategy Devices





Thanks you.

For questions, please contact :

yhkim@hanaro.com